

2.6 Independence

Definition

Two events A and B are **independent**, if $P(A|B) = P(A)$ or $P(B|A) = P(B)$.

Corollary

- If A and B are independent, then $P(A \cap B) = P(A) \cdot P(B)$.

This follows immediately from probability rule R.4.1.

- If A and B are independent, then the same is true for A' and B' .

Proof

$$\begin{aligned}
 P(A' \cap B') &= P((A \cup B)') \\
 &= 1 - P(A \cup B) \\
 &= 1 - P(A) - P(B) + P(A \cap B) \\
 &= 1 - P(A) - P(B) + P(A) \cdot P(B) \\
 &= (1 - P(A)) \cdot 1 - (1 - P(A)) \cdot P(B) \\
 &= (1 - P(A)) \cdot (1 - P(B)) = P(A') \cdot P(B')
 \end{aligned}$$

Examples

- See the semiconductor manufacturing in Example 3.

$$\left. \begin{aligned} P(B) &= 0.024 \\ P(B|A) &= 0.1 \end{aligned} \right\} \Rightarrow A \text{ and } B \text{ are dependent}$$

- Given is a circuit with three devices A , B and C . Assume that devices fail independently. $P(A)$ = the probability that A functions = 0.95. Analogously, $P(B) = 0.95$ and $P(C) = 0.90$.

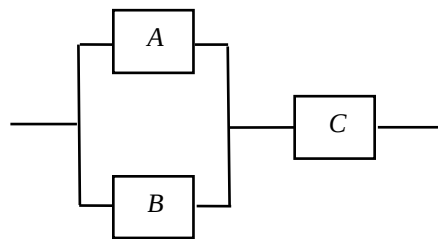


Figure 2.5 Circuit with three devices

$$\begin{aligned}
 P(\text{circuit functions}) &= P((A \cup B) \cap C) \\
 &= P(A \cup B) \cdot P(C)
 \end{aligned}$$

Independency

$$\begin{aligned}
 P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\
 &= P(A) + P(B) - P(A) \cdot P(B)
 \end{aligned}$$

Independency

$$= 0.95 + 0.95 - 0.95^2 = 0.9975$$

Alternative method for the left part of the circuit, which contains two **parallel** devices:

$$\begin{aligned} P(A \cup B) &= 1 - P((A \cup B)') \\ &= 1 - P(A' \cap B') \\ &= 1 - P(A') \cdot P(B') \\ &= 1 - (1 - P(A)) \cdot (1 - P(B)) \\ &= 1 - 0.05^2 = 0.9975 \end{aligned}$$

$$\text{So, } P(\text{circuit functions}) = 0.9975 \cdot 0.9 = 0.8978$$

2.7 Bayes' Theorem

Suppose the semiconductor from Example 3 fails ($= B$). What is the probability that the chip was subjected to high levels of contamination during manufacturing ($= A$)?

So,

$$P(A|B) = ?$$

If there is an **unnatural order** when calculating a conditional probability, the order of A and B must be changed. This is done using **Bayes' Theorem**.

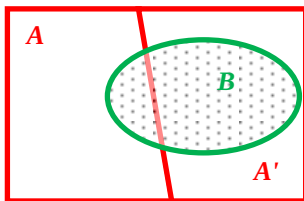


Figure 2.6 Bayes' Theorem

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Proof

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A) \cdot P(A)}{P(B)}$$

R4.2

Example 6

See the semiconductor manufacturing in Example 3.

$$\begin{aligned} P(\text{high levels of contamination} \mid \text{semiconductor fails}) &= P(A|B) \\ &= \frac{P(B|A) \cdot P(A)}{P(B)} \\ &= \frac{0.1 \cdot 0.2}{0.024} \\ &= 0.8333 \end{aligned}$$

Bayes

2.8 Random Variables

Given is a random experiment E (e.g. ruling out a target in a maximum of five attempts) with a corresponding sample space S

$$S = \{ h, mh, mmh, mmmh, mmmmh \}$$

where m = miss and h = hit.

A function X that assigns a real number to each outcome in the sample space S is called a random variable. A discrete random variable is a random variable with a finite or countably infinite range. All other random variables are called continuous. So the range of a continuous random variable must be an interval (either finite or infinite) of real numbers.

Examples

7. Ruling out a target in a maximum of five attempts. Let X = the number of the attempt at which the target is ruled out.
So $X(mmh) = 3$ and $X(h) = 1$. Since the range of $X = \{1, 2, 3, 4, 5\}$, X has a finite range. So X is a discrete random variable.
8. Determining the life length of a light bulb. Let X = the life length of the light bulb. Since the range of $X = \{x \in \mathbb{R} \mid x \geq 0\}$, X is a continuous random variable.
9. Ruling out a target in an infinite number of independent attempts with constant probability p of success. Let X = the number of the attempt at which the target is ruled out. Since the range of $X = \{1, 2, 3, \dots\}$, X has a countably infinite range. So X is a discrete random variable.